THE SHEAF OF RELATIVE DIFFERENTIALS IS CANONICALLY ISOMORPHIC TO THE CONORMAL SHEAF OF THE DIAGONAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-differentials-diagonal is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 6506-6511 and proof span 6513-6595 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof cites two labels of the same chapter (the conormal sheaf definition and the affine conormal lemma) and several labels of other chapters (Schemes, Modules, Algebra); all of them are delivered as external navigation pointers only.

Editorial note. 6 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The diagonal morphism is an immersion, so that it has an ideal sheaf on an open subscheme of the fibre product containing its image; the author invokes the same-chapter statement for this and for the shape of the chosen open. Invoked at proof lines 6515-6519 and again at 6551-6554. Supplies the open W and the ideal sheaf whose square-zero thickening carries the two sections.
- The conormal sheaf of an immersion, its definition, and its description over an affine chart as I/I^2, together with the affine description of the module of differentials. Definition and lemmas of the same chapter cited at proof lines 6526-6528 and 6572-6576. Names the target of the canonical map and lets the affine computation identify both sides with J/J^2 and with the module of differentials of the coordinate ring.
- The double-structure mechanism: a ring surjection A to B carrying two sections gives a derivation B to the kernel of the surjection, and the universal property of the module of differentials turns it into a linear map. Cited as an external chapter lemma at proof lines 6530 and 6540, and applied to the

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square-zero thickening O_W/J^2 at proof lines 6540-6548. Produces the canonical map from the module of differentials to the conormal sheaf.

Source reference. Original reference algebra-lemma-differentials-diagonal: sources/stacks-morphisms/pinned-original-source.tex, source lines 6594–6594. Same-chapter reference used in the retained author text at source line(s) 6594. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original definition definition-conormal-sheaf: sources/stacks-morphisms/pinned-original-source.tex, source lines 6164–6164. Same-chapter reference used in the retained author text at source line(s) 6537. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma lemma-affine-conormal: sources/stacks-morphisms/pinned-original-source.tex, source lines 6186–6186. Same-chapter reference used in the retained author text at source line(s) 6538, 6588. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma lemma-differentials-affine: sources/stacks-morphisms/pinned-original-source.tex, source lines 6441–6441. Same-chapter reference used in the retained author text at source line(s) 6495, 6590, 6701, 6716, 7660. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original reference modules-lemma-double-structure-gives-derivation: sources/stacks-morphisms/pinned-original-source.tex, source lines 6546–6546. Same-chapter reference used in the retained author text at source line(s) 6546, 6559. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original reference schemes-lemma-diagonal-immersion: sources/stacks-morphisms/pinned-original-source.tex, source lines 6520–6520. Same-chapter reference used in the retained author text at source line(s) 6520, 6525, 6571. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $f : X \to S$ be a morphism of schemes. There is a canonical isomorphism between $\Omega_{X/S}$ and the conormal sheaf of the diagonal morphism $\Delta_{X/S} : X \longrightarrow X \times_S X$.*

Proof. We first establish the existence of a couple of "global" sheaves and global maps of sheaves, and further down we describe the constructions over some affine opens.

Recall that $\Delta = \Delta_{X/S} : X \to X \times_S X$ is an immersion, see Schemes, Lemma schemes-lemma-diagonal-immersion. Let $\mathcal{J}$ be the ideal sheaf of the immersion which lives over some open subscheme W of $X \times_S X$ such that $\Delta(X) \subset W$ is closed.

Let us take the one that was found in the proof of Schemes, Lemma schemes-lemma-diagonal-immersion. Note that the sheaf of rings $\mathcal{O}_W/\mathcal{J}^2$ is supported on $\Delta(X)$. Moreover it sits in a short exact sequence of sheaves

$$0 \to \mathcal{J}/\mathcal{J}^2 \to \mathcal{O}_W/\mathcal{J}^2 \to \Delta_*\mathcal{O}_X \to 0.$$

Using Δ^{-1} we can think of this as a surjection of sheaves of $f^{-1}\mathcal{O}_S$-algebras with kernel the conormal sheaf of Δ (see Definition definition-conormal-sheaf and Lemma lemma-affine-conormal).

$$0 \to \mathcal{C}_{X/X\times_S X} \to \Delta^{-1}(\mathcal{O}_W/\mathcal{J}^2) \to \mathcal{O}_X \to 0$$

This places us in the situation of Modules, Lemma modules-lemma-double-structure-gives-derivation. The projection morphisms $p_i : X \times_S X \to X$, $i = 1, 2$ induce maps of sheaves of rings $(p_i)^\sharp : (p_i)^{-1}\mathcal{O}_X \to \mathcal{O}_{X\times_S X}$. We may restrict to W and quotient by $\mathcal{J}^2$ to get $(p_i)^{-1}\mathcal{O}_X \to \mathcal{O}_W/\mathcal{J}^2$. Since $\Delta^{-1}p_i^{-1}\mathcal{O}_X = \mathcal{O}_X$ we get maps

$$s_i : \mathcal{O}_X \to \Delta^{-1}(\mathcal{O}_W/\mathcal{J}^2).$$

Both s_1 and s_2 are sections to the map $\Delta^{-1}(\mathcal{O}_W/\mathcal{J}^2) \to \mathcal{O}_X$, as in Modules, Lemma modules-lemma-double-structure-gives-derivation. Thus we get an S-derivation $\mathrm{d} = s_2 - s_1 : \mathcal{O}_X \to \mathcal{C}_{X/X\times_S X}$. By the universal property of the module of differentials we find a unique $\mathcal{O}_X$-linear map

$$\Omega_{X/S} \longrightarrow \mathcal{C}_{X/X\times_S X}, \quad f\mathrm{d}g \longmapsto fs_2(g) - fs_1(g)$$

To see the map is an isomorphism, let us work this out over suitable affine opens. We can cover X by affine opens $\mathrm{Spec}(A) = U \subset X$ whose image is contained in an affine open $\mathrm{Spec}(R) = V \subset S$. According to the proof of Schemes, Lemma schemes-lemma-diagonal-immersion $U \times_V U \subset X \times_S X$ is an affine open contained in the open W mentioned above. Also $U \times_V U = \mathrm{Spec}(A \otimes_R A)$. The sheaf $\mathcal{J}$ corresponds to the ideal $J = \mathrm{Ker}(A \otimes_R A \to A)$. The short exact sequence to the short exact sequence of $A \otimes_R A$-modules

$$0 \to J/J^2 \to (A \otimes_R A)/J^2 \to A \to 0$$

The sections s_i correspond to the ring maps

$$A \longrightarrow (A \otimes_R A)/J^2, \quad s_1 : a \mapsto a \otimes 1, \quad s_2 : a \mapsto 1 \otimes a.$$

By Lemma lemma-affine-conormal we have $\Gamma(U, \mathcal{C}_{X/X\times_S X}) = J/J^2$ and by Lemma lemma-differentials-affine we have $\Gamma(U, \Omega_{X/S}) = \Omega_{A/R}$. The map above is the map $a\mathrm{d}b \mapsto a \otimes b - ab \otimes 1$ which is shown to be an isomorphism in Algebra, Lemma algebra-lemma-differentials-diagonal. $\square$